Muhammad Hassan

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Professional Summary

AI Automation Engineer and Python Developer with hands-on experience building end-to-end intelligent systems using LLMs, LangChain, RAG pipelines, and ML. Delivered deployed, production-ready projects including AI chatbots, automated data pipelines, and real-time ML dashboards. Strong foundation in OOP, data structures, algorithms, and software engineering principles. Eager to contribute to large-scale AI infrastructure and ML systems at Google.

Technical Skills

Languages: Python, SQL, Bash, LaTeX
AI / ML: LangChain, RAG Pipelines, Ollama (LLaMA 3), OpenAI API, Scikit-learn, XGBoost, Random Forest, NumPy, Pandas
Frameworks & Tools: FastAPI, Streamlit, n8n (AI Workflow Automation), Flowise, BeautifulSoup4, Selenium
Data & Visualization: Matplotlib, Seaborn, Jupyter Notebook, MySQL
Dev & Deployment: Git, GitHub, VS Code, Pydroid 3, Linux CLI
Concepts: OOP (4 Pillars), REST APIs, Vector Databases, AI Agents, Decorators, Generators, Exception Handling

Projects

RAG-Based PDF Chatbot | *Python, LangChain, FAISS, Ollama, Streamlit* github.com/mhassan619/AI_Automation_Engineer
- Built a Retrieval-Augmented Generation (RAG) pipeline that ingests PDF documents, chunks and embeds text into a FAISS vector store, and retrieves context-aware answers using a locally hosted LLaMA 3 model via Ollama.
- Designed an interactive Streamlit UI enabling real-time Q&A over arbitrary PDFs with zero cloud dependency.
- Reduced hallucination rate by grounding responses exclusively in retrieved document context.

LangChain Study Assistant | *Python, LangChain, OpenAI API, Streamlit*
- Developed a conversational AI study assistant using LangChain's memory and chain abstractions, enabling multi-turn dialogues with persistent context across sessions.
- Integrated prompt engineering patterns (system prompts, few-shot examples) to improve answer accuracy and relevance.

ML Analytics Dashboard | *Python, Scikit-learn, XGBoost, Streamlit, Pandas* Live Demo
- Built and deployed an interactive ML dashboard showcasing real-time model predictions, feature importance, and evaluation metrics (accuracy, ROC-AUC) using Scikit-learn and XGBoost classifiers.
- Implemented data preprocessing pipelines including missing value imputation, label encoding, and train/test splitting.

Ollama CLI AI Chatbot | *Python, Ollama, LLaMA 3, REST API*
- Engineered a lightweight command-line chatbot interfacing with Ollama's local REST API to run LLaMA 3 entirely on-device, achieving private, offline inference with sub-second response times.

AI Automation Dashboard | *n8n, Python, Streamlit, REST APIs* Live Demo
- Designed multi-step automation workflows in n8n (self-hosted) connecting APIs, web scrapers, and notification services to eliminate repetitive manual tasks for small business use cases.

Goat Farm OOP Management System | *Python, OOP, File I/O*
- Architected a domain-specific Python application demonstrating all four OOP pillars — encapsulation, inheritance, polymorphism, and abstraction — across a modular class hierarchy for livestock record management.

Education

University of Layyah Layyah, Punjab, Pakistan
Bachelor of Science in Computer Science *Oct 2024 – Jun 2028*
- Relevant Coursework: Object-Oriented Programming, Data Structures, Linear Algebra, Digital Logic Design, Applied Physics, Database Systems

Certifications

Kaggle: Intro to Programming, Intro to Machine Learning, Intermediate Machine Learning, Python Machine Learning, Matplotlib

Achievements & Activities

Open Source Portfolio — AI Automation Engineer Curriculum
- Completed a self-directed 24-day intensive curriculum covering advanced Python, Git/GitHub, LangChain, RAG, and AI agents; all projects documented and published on GitHub.
- Solved algorithmic problems on LeetCode (Two Sum, Add Two Numbers) in Python, strengthening data structures and problem-solving skills relevant to coding interviews.
- Independently built and deployed two live Streamlit applications publicly accessible on the internet.